Mathematics Class 6 — Solved Exercises: Unit 1, Sub-Domain 1 (Factors and Multiples) — Complete Sub-Domain

Mathematics (SNC 2023-24) — English Medium — Class 6 — Punjab Curriculum and Textbook Board (PCTB)

Unit 1 — Numbers and Operations: Sub-Domain 1, Factors and Multiples

Practice 1 — List all the factors of each number (book page 2)

(i) Factors of 84.

Step 1: Write 84 as products of two whole numbers: 1 x 84, 2 x 42, 3 x 28, 4 x 21, 6 x 14, 7 x 12. Step 2: Collect every number used: 1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42, 84. Step 3: Stop here because the pairs now start repeating. Final answer: the factors of 84 are 1, 2, 3, 4, 6, 7, 12, 14, 21, 28, 42 and 84.

(ii) Factors of 120.

Step 1: Factor pairs: 1 x 120, 2 x 60, 3 x 40, 4 x 30, 5 x 24, 6 x 20, 8 x 15, 10 x 12. Step 2: Collect them in order: 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 20, 24, 30, 40, 60, 120. Final answer: the factors of 120 are 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 20, 24, 30, 40, 60 and 120.

(iii) Factors of 230.

Step 1: Factor pairs: 1 x 230, 2 x 115, 5 x 46, 10 x 23. Step 2: Collect them in order: 1, 2, 5, 10, 23, 46, 115, 230. Final answer: the factors of 230 are 1, 2, 5, 10, 23, 46, 115 and 230.

(iv) Factors of 352.

Step 1: Note 352 = 32 x 11. Factor pairs: 1 x 352, 2 x 176, 4 x 88, 8 x 44, 11 x 32, 16 x 22. Step 2: Collect them in order: 1, 2, 4, 8, 11, 16, 22, 32, 44, 88,

176, 352. Final answer: the factors of 352 are 1, 2, 4, 8, 11, 16, 22, 32, 44, 88, 176 and 352.

Practice 2 — List the first five multiples of each number (book page 3)

(i) First five multiples of 3.

Step 1: Multiply 3 by 1, 2, 3, 4 and 5. Step 2: 1 x 3 = 3, 2 x 3 = 6, 3 x 3 = 9, 4 x 3 = 12, 5 x 3 = 15. Final answer: 3, 6, 9, 12, 15.

(ii) First five multiples of 11.

Step 1: Multiply 11 by 1, 2, 3, 4 and 5. Step 2: 11, 22, 33, 44, 55. Final answer: 11, 22, 33, 44, 55.

(iii) First five multiples of 29.

Step 1: Multiply 29 by 1, 2, 3, 4 and 5. Step 2: 29, 58, 87, 116, 145. Final answer: 29, 58, 87, 116, 145.

(iv) First five multiples of 73.

Step 1: Multiply 73 by 1, 2, 3, 4 and 5. Step 2: 73, 146, 219, 292, 365. Final answer: 73, 146, 219, 292, 365.

Practice 3 — Write at least 5 multiples and all factors of each number (book page 4)

(i) 36.

Multiples: Step 1: Multiply 36 by 1, 2, 3, 4 and 5 to get 36, 72, 108, 144, 180.

Factors: Step 2: Factor pairs are 1 x 36, 2 x 18, 3 x 12, 4 x 9, 6 x 6. Step 3: Hence the factors are 1, 2, 3, 4, 6, 9, 12, 18 and 36.

(ii) 45.

Multiples: Step 1: Multiply 45 by 1, 2, 3, 4 and 5 to get 45, 90, 135, 180, 225.

Factors: Step 2: Factor pairs are 1 x 45, 3 x 15, 5 x 9. Step 3: Hence the factors are 1, 3, 5, 9, 15 and 45.

(iii) 76.

Multiples: Step 1: Multiply 76 by 1, 2, 3, 4 and 5 to get 76, 152, 228, 304, 380.

Factors: Step 2: Factor pairs are 1 x 76, 2 x 38, 4 x 19. Step 3: Hence the factors are 1, 2, 4, 19, 38 and 76.

Practice 4 — Twin primes between 1 and 100 (book page 6)

Twin primes are prime numbers that differ by 2, such as 3 and 5. List all the twin primes between 1 and 100.

Step 1: List the primes from 1 to 100: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97. Step 2: Pick out pairs that differ by 2. Step 3: The pairs are (3, 5), (5, 7), (11, 13), (17, 19), (29, 31), (41, 43), (59, 61) and (71, 73). Final answer: 8 twin-prime pairs.

Practice 5 — Find the prime factors (book page 7)

(i) Prime factors of 30.

Step 1: Divide by the smallest prime: 30 / 2 = 15. Step 2: 15 / 3 = 5. Step 3: 5 is prime, so stop. Final answer: 30 = 2 x 3 x 5.

(ii) Prime factors of 210.

Step 1: 210 / 2 = 105. Step 2: 105 / 3 = 35. Step 3: 35 / 5 = 7. Step 4: 7 is prime, so stop. Final answer: 210 = 2 x 3 x 5 x 7.

(iii) Prime factors of 633.

Step 1: 633 is odd, and 6 + 3 + 3 = 12, which is divisible by 3, so 633 / 3 = 211. Step 2: 211 has no divisor other than 1 and itself, so it is prime. Final answer: 633 = 3 x 211.

(iv) Prime factors of 310.

Step 1: 310 / 2 = 155. Step 2: 155 / 5 = 31. Step 3: 31 is prime, so stop. Final answer: 310 = 2 x 5 x 31.

Exercise 1.1 (book page 6)

1. Write all the factors of: (i) 30 (ii) 41 (iii) 65 (iv) 78 (v) 105 (vi) 118 (vii) 165 (viii) 220.

(i) 30: 1, 2, 3, 5, 6, 10, 15, 30. (ii) 41: 1, 41 (41 is prime). (iii) 65: 1, 5, 13, 65. (iv) 78: 1, 2, 3, 6, 13, 26, 39, 78. (v) 105: 1, 3, 5, 7, 15, 21, 35, 105. (vi) 118: 1, 2, 59, 118. (vii) 165: 1, 3, 5, 11, 15, 33, 55, 165. (viii) 220: 1, 2, 4, 5, 10, 11, 20, 22, 44, 55, 110, 220. Each list was found by pairing factors (1 x n, then the next divisor) until the pairs repeat.

2. Write the first five multiples of: (i) 7 (ii) 13 (iii) 19 (iv) 43 (v) 59.

(i) 7: 7, 14, 21, 28, 35. (ii) 13: 13, 26, 39, 52, 65. (iii) 19: 19, 38, 57, 76, 95. (iv) 43: 43, 86, 129, 172, 215. (v) 59: 59, 118, 177, 236, 295. Each list is the number multiplied by 1, 2, 3, 4 and 5 in turn.

3. Write all prime numbers between 30 and 65.

Step 1: Test each number from 31 to 64 for divisors other than 1 and itself. Step 2: The numbers with exactly two factors are 31, 37, 41, 43, 47, 53, 59 and 61. Final answer: 31, 37, 41, 43, 47, 53, 59, 61.

4. Write all even prime numbers between 50 and 100.

Step 1: The only even prime number is 2, because every other even number has 2 as a factor besides 1 and itself. Step 2: 2 is not between 50 and 100. Final answer: there is no even prime number between 50 and 100.

5. List down all even prime natural and whole numbers.

Step 1: A prime number has exactly two factors, 1 and itself. Step 2: Among even numbers, only 2 has exactly two factors; every larger even number is

also divisible by 2. Final answer: 2 is the only even prime number, in both natural and whole numbers.

6. List down all odd composite numbers between 50 and 100.

Step 1: A composite number has more than two factors, and an odd number is not divisible by 2. Step 2: Checking the odd numbers from 51 to 99 gives 51, 55, 57, 63, 65, 69, 75, 77, 81, 85, 87, 91, 93, 95 and 99. Final answer: 51, 55, 57, 63, 65, 69, 75, 77, 81, 85, 87, 91, 93, 95, 99.

7. Write all neither prime nor composite natural and whole numbers.

Step 1: 1 has only one factor (itself), so it is neither prime nor composite. Step 2: In whole numbers, 0 is also neither prime nor composite. Final answer: natural numbers: 1; whole numbers: 0 and 1.

Practice 6 — Prime factors in index notation (book page 8)

(i) 96.

Step 1: Divide by 2 repeatedly: 96 / 2 = 48. Step 2: 48 / 2 = 24. Step 3: 24 / 2 = 12. Step 4: 12 / 2 = 6. Step 5: 6 / 2 = 3. Step 6: 3 is prime, so stop. Final answer: 96 = 2 x 2 x 2 x 2 x 2 x 3 = 2^5 x 3.

(ii) 192.

Step 1: 192 / 2 = 96. Step 2: 96 / 2 = 48. Step 3: 48 / 2 = 24. Step 4: 24 / 2 = 12. Step 5: 12 / 2 = 6. Step 6: 6 / 2 = 3. Step 7: 3 is prime, so stop. Final answer: 192 = 2^6 x 3.

(iii) 576.

Step 1: 576 / 2 = 288. Step 2: 288 / 2 = 144. Step 3: 144 / 2 = 72. Step 4: 72 / 2 = 36. Step 5: 36 / 2 = 18. Step 6: 18 / 2 = 9. Step 7: 9 / 3 = 3. Step 8: 3 / 3 = 1. Final answer: 576 = 2^6 x 3^2.

(iv) 4,032.

Step 1: 4,032 / 2 = 2,016. Step 2: 2,016 / 2 = 1,008. Step 3: 1,008 / 2 = 504. Step 4: 504 / 2 = 252. Step 5: 252 / 2 = 126. Step 6: 126 / 2 = 63. Step 7: 63 / 3 = 21. Step 8: 21 / 3 = 7. Step 9: 7 / 7 = 1. Final answer: 4,032 = 2^6 x 3^2 x 7.

Challenge (book page 8): I am the product of two different prime factors. One of my factors is raised to the 2nd power. My value is between 62 and 82. Who am I?

Step 1: Try a squared prime times a different prime. Step 2: 2^2 x 17 = 4 x 17 = 68, which lies between 62 and 82. Final answer: one such number is 68.

Note: the wording as printed admits more than one value (for example 63 = 3^2 x 7, 75 = 5^2 x 3 and 76 = 2^2 x 19 also fit), so 68 is a valid example, not the only value.

Exercise 1.2 (book pages 8-9)

1. Fill the spaces to complete the factor tree: (i) 20 (ii) 105 (iii) 180.

Step 1: (i) 20 = 4 x 5, and 4 = 2 x 2, so the leaves are 2, 2 and 5. Final answer: 20 = 2^2 x 5. Step 2: (ii) 105 = 15 x 7, and 15 = 3 x 5, so the leaves are 3, 5 and 7. Final answer: 105 = 3 x 5 x 7. Step 3: (iii) 180 = 18 x 10 = (2 x 3 x 3) x (2 x 5), so the leaves are 2, 2, 3, 3 and 5. Final answer: 180 = 2^2 x 3^2 x 5.

2. Write the following expanded forms into index notation: (i) 5 x 5 x 5 x 5 x 7 (ii) 2 x 2 x 5 x 5 x 5 x 5 x 7 x 7 x 7 (iii) 3 x 3 x 3 x 3 x 5 x 5 x 5 (iv) 11 x 11 x 11 x 11 x 13 x 19 x 19 x 19.

Step 1: Count each repeated factor. (i) 5 occurs 4 times and 7 once: 5^4 x 7. Step 2: (ii) 2 occurs 2 times, 5 occurs 4 times and 7 occurs 3 times: 2^2 x 5^4 x 7^3. Step 3: (iii) 3 occurs 4 times and 5 occurs 3 times: 3^4 x 5^3. Step 4: (iv) 11 occurs 4 times, 13 once and 19 three times: 11^4 x 13 x 19^3.

Final answers: (i) 5^4 x 7, (ii) 2^2 x 5^4 x 7^3, (iii) 3^4 x 5^3, (iv) 11^4 x 13 x 19^3.

3. Find the prime factors for the following using division method then express in index notation: (i) 32 (ii) 90 (iii) 110 (iv) 115 (v) 225 (vi) 540 (vii) 1386 (viii) 2205 (ix) 6615 (x) 6075.

Step 1: Divide each number by the smallest prime until 1 is reached, then count repeats. (i) 32 = 2^5. (ii) 90 = 2 x 3^2 x 5. (iii) 110 = 2 x 5 x 11. (iv) 115 = 5 x 23. (v) 225 = 3^2 x 5^2. (vi) 540 = 2^2 x 3^3 x 5. (vii) 1386 = 2 x 3^2 x 7 x 11. (viii) 2205 = 3^2 x 5 x 7^2. (ix) 6615 = 3^3 x 5 x 7^2. (x) 6075 = 3^5 x 5^2. Each result was checked by dividing step by step until the quotient was 1.

4. Find the prime factors of the following numbers using factor tree method then express in index notation: (i) 64 (ii) 125 (iii) 100 (iv) 500 (v) 188 (vi) 648 (vii) 1512 (viii) 3240 (ix) 7560 (x) 7875.

Step 1: Split each number into two factors, then split again until only primes remain. (i) 64 = 2^6. (ii) 125 = 5^3. (iii) 100 = 2^2 x 5^2. (iv) 500 = 2^2 x 5^3. (v) 188 = 2^2 x 47. (vi) 648 = 2^3 x 3^4. (vii) 1512 = 2^3 x 3^3 x 7. (viii) 3240 = 2^3 x 3^4 x 5. (ix) 7560 = 2^3 x 3^3 x 5 x 7. (x) 7875 = 3^2 x 5^3 x 7.

5. Prove that prime factors of 2,600 are same using division method and factor tree method.

Step 1: Division method: 2,600 / 2 = 1,300. Step 2: 1,300 / 2 = 650. Step 3: 650 / 2 = 325. Step 4: 325 / 5 = 65. Step 5: 65 / 5 = 13. Step 6: 13 / 13 = 1. So 2,600 = 2^3 x 5^2 x 13. Step 7: Factor tree method: 2,600 = 26 x 100 = (2 x 13) x (4 x 25) = 2 x 13 x 2 x 2 x 5 x 5, the same primes. Final answer: both methods give 2,600 = 2^3 x 5^2 x 13, so the prime factors are the same.

6. I am a product of three different numbers. My greatest prime factor is 13. I am between 100 and 200. Who am I?

Step 1: Pick three different factors whose greatest prime factor is 13, for example 2, 5 and 13. Step 2: Multiply: 2 x 5 x 13 = 130, which lies between 100 and 200. Final answer: one such number is 130. Note: the wording as printed admits more than one value (for example 182 = 2 x 7 x 13 and 195 = 3 x 5 x 13 also fit), so 130 is a valid example, not the only value.

7. Find the number that is the product of the most prime numbers below 100.

Step 1: The wording as printed does not fix a single number, so two exact facts are given. Step 2: The greatest prime number below 100 is 97. Step 3: The product of ALL the prime numbers below 100 (2, 3, 5, ..., 97) is 2305567963945518424753102147331756070, a 37-digit number. Final answer: if the question means the product of all primes below 100, it is 2305567963945518424753102147331756070; if it means the greatest prime below 100, it is 97.

8. Write these index notations into expanded form: (i) 2^4 x 3^2 x 5^3 (ii) 3^5 x 5^4 x 11^3 (iii) 5^6 x 7^3 x 11^5 (iv) 2^7 x 3^4 x 13^3 x 17^2.

Step 1: (i) 2^4 = 16, 3^2 = 9, 5^3 = 125; 16 x 9 x 125 = 18,000. Step 2: (ii) 3^5 = 243, 5^4 = 625, 11^3 = 1,331; 243 x 625 x 1,331 = 202,145,625. Step 3: (iii) 5^6 = 15,625, 7^3 = 343, 11^5 = 161,051; the product is 863,132,703,125. Step 4: (iv) 2^7 = 128, 3^4 = 81, 13^3 = 2,197, 17^2 = 289; the product is 6,582,985,344. Final answers: (i) 18,000, (ii) 202,145,625, (iii) 863,132,703,125, (iv) 6,582,985,344.

Practice 7 — Greatest length of measuring tape for 160 m, 280 m and 200 m (book page 12)

Find the greatest length of measuring tape which can be used to measure exactly 160 m, 280 m and 200 m length.

Step 1: The tape length must divide all three lengths, so find H.C.F of 160, 280 and 200. Step 2: 160 = 2^5 x 5, 280 = 2^3 x 5 x 7, 200 = 2^3 x 5^2. Step 3: The common prime factors are 2^3 and 5. Step 4: H.C.F = 2^3 x 5 = 40. Final answer: the greatest length of measuring tape is 40 m.

Exercise 1.3 — H.C.F (book page 12)

1. Find H.C.F of each set of numbers using prime factorization method: (i) 6 and 9 (ii) 25 and 45 (iii) 18 and 28 (iv) 5, 30 and 60 (v) 14, 28 and 112 (vi) 36, 54 and 108.

Step 1: Factorize each set and take the product of common prime factors. (i) 6 = 2 x 3, 9 = 3^2, common 3, so H.C.F = 3. (ii) 25 = 5^2, 45 = 3^2 x 5, common 5, so H.C.F = 5. (iii) 18 = 2 x 3^2, 28 = 2^2 x 7, common 2, so H.C.F = 2. (iv) 5 = 5, 30 = 2 x 3 x 5, 60 = 2^2 x 3 x 5, common 5, so H.C.F = 5. (v) 14 = 2 x 7, 28 = 2^2 x 7, 112 = 2^4 x 7, common 2 x 7 = 14, so H.C.F = 14. (vi) 36 = 2^2 x 3^2, 54 = 2 x 3^3, 108 = 2^2 x 3^3, common 2 x 3^2 = 18, so H.C.F = 18.

2. Find H.C.F of each set of numbers using long division method: (i) 110 and 170 (ii) 104 and 234 (iii) 400 and 3996 (iv) 32, 96 and 256 (v) 30, 168 and 288 (vi) 80, 215, 245 and 720.

Step 1: Divide the larger number by the smaller, then divide the divisor by the remainder, until the remainder is zero; the last divisor is the H.C.F. (i) 170 / 110 leaves 60, 110 / 60 leaves 50, 60 / 50 leaves 10, 50 / 10 leaves 0, so H.C.F = 10. (ii) 234 / 104 leaves 26, 104 / 26 leaves 0, so H.C.F = 26. (iii)

3,996 / 400 leaves 396, 400 / 396 leaves 4, 396 / 4 leaves 0, so H.C.F = 4. (iv) H.C.F of 32 and 96 is 32, and H.C.F of 32 and 256 is 32, so H.C.F = 32. (v) H.C.F of 30 and 168 is 6, and H.C.F of 6 and 288 is 6, so H.C.F = 6. (vi) H.C.F of 80 and 215 is 5, of 5 and 245 is 5, and of 5 and 720 is 5, so H.C.F = 5.

3. What is H.C.F of any two prime numbers?

Step 1: Two different prime numbers have no common factor except 1. Step 2: Therefore their highest common factor is 1, and such numbers are called co-prime. Final answer: the H.C.F of any two (different) prime numbers is 1.

4. Can the highest common factor of 8 and 64 be greater than 8? Explain your reasoning.

Step 1: The H.C.F is always smaller than or equal to the smallest number in the set. Step 2: Here the smallest number is 8, so the H.C.F cannot be greater than 8. Step 3: In fact, 8 divides 64 exactly (64 / 8 = 8), so the H.C.F of 8 and 64 is 8 itself. Final answer: No, it cannot be greater than 8.

5. Aleezey has 16 Science books, 28 Mathematics books and 30 English books. She wants to pack these books equally without mixing the books and with no left over. What is the greatest number of books she can put in each package?

Step 1: The number in each package must divide 16, 28 and 30, so find their H.C.F. Step 2: 16 = 2^4, 28 = 2^2 x 7, 30 = 2 x 3 x 5. Step 3: The only common prime factor is 2. Final answer: the greatest number of books in each package is 2.

6. There are 50 students of grade VI, 70 students of grade VII and 80 students of grade VIII. A teacher wishes to arrange them in equal number of students in each row. Find the greatest number of students that could be arranged in a row.

Step 1: Find H.C.F of 50, 70 and 80. Step 2: 50 = 2 x 5^2, 70 = 2 x 5 x 7, 80 = 2^4 x 5. Step 3: Common factors are 2 x 5 = 10. Final answer: the greatest number of students in each row is 10.

7. A room is 10 m 15 cm long and 8 m 80 cm wide. The floor of the room is to be paved with square tiles. Find the length of the largest tile that can be used.

Step 1: Convert to centimetres: 10 m 15 cm = 1,015 cm, 8 m 80 cm = 880 cm. Step 2: Find H.C.F of 1,015 and 880. Step 3: 1,015 = 5 x 7 x 29, 880 = 2^4 x 5 x 11. Step 4: The only common factor is 5. Final answer: the length of the largest square tile is 5 cm.

Practice 8 — Least number of trees in rows of 7, 9 and 15 (book page 14)

Waqas is planting trees. He has enough trees to plant 7, 9 and 15 trees in each row. What is the least number of trees Waqas could have?

Step 1: The number of trees must be a multiple of 7, 9 and 15, so find their L.C.M. Step 2: 7 = 7, 9 = 3^2, 15 = 3 x 5. Step 3: L.C.M = 3^2 x 5 x 7 = 315. Final answer: the least number of trees is 315.

Exercise 1.4 — L.C.M (book pages 14-15)

1. Find L.C.M of each set of numbers using prime factorization method: (i) 8 and 18 (ii) 28 and 49 (iii) 20, 90 and 180 (iv) 25, 35 and 45 (v) 15, 25 and 65 (vi) 150, 250 and 350.

Step 1: L.C.M = product of common prime factors x product of non-common prime factors. (i) 8 = 2^3, 18 = 2 x 3^2, so L.C.M = 2^3 x 3^2 = 72. (ii) 28 = 2^2 x 7, 49 = 7^2, so L.C.M = 2^2 x 7^2 = 196. (iii) 180 is already a multiple of 20 and 90, so L.C.M = 180. (iv) 25 = 5^2, 35 = 5 x 7, 45 = 3^2 x 5, so L.C.M = 3^2 x 5^2 x 7 = 1,575. (v) 15 = 3 x 5, 25 = 5^2, 65 = 5 x 13,

so L.C.M = 3 x 5^2 x 13 = 975. (vi) 150 = 2 x 3 x 5^2, 250 = 2 x 5^3, 350 = 2 x 5^2 x 7, so L.C.M = 2 x 3 x 5^3 x 7 = 5,250.

2. Find L.C.M of each set of numbers using division method: (i) 14 and 63 (ii) 25 and 45 (iii) 14, 28 and 84 (iv) 7, 28 and 98 (v) 220, 240, 260 and 280.

Step 1: Use the division method: divide by the smallest prime that divides at least one number, bring down numbers not divided exactly, and continue until the last row is co-prime; L.C.M is the product of all divisors. (i) L.C.M of 14 and 63 = 126. (ii) L.C.M of 25 and 45 = 225. (iii) 84 is already a multiple of 14 and 28, so L.C.M = 84. (iv) L.C.M of 7, 28 and 98 = 196. (v) L.C.M of 220, 240, 260 and 280 = 240,240.

3. A frog and a grasshopper start jumping together and jump along the same path. The frog always jumps 25 cm and the grasshopper always jumps 15 cm. At what distance they will meet again?

Step 1: They meet again at a common multiple of 25 and 15. Step 2: L.C.M of 25 and 15 = 75. Final answer: they will meet again at 75 cm.

4. A university bus arrives near UET Lahore in every 280 seconds and an orange train arrives in every 360 seconds. How often do train and bus arrive at the same time?

Step 1: Find L.C.M of 280 and 360. Step 2: 280 = 2^3 x 5 x 7, 360 = 2^3 x 3^2 x 5. Step 3: L.C.M = 2^3 x 3^2 x 5 x 7 = 2,520 seconds. Step 4: 2,520 seconds = 42 minutes. Final answer: the bus and the train arrive together every 2,520 seconds, that is, every 42 minutes.

5. I am thinking of a number that is divisible by 29, 58 and 116. What is the smallest possible number that I could be thinking of?

Step 1: Find L.C.M of 29, 58 and 116. Step 2: 116 is already a multiple of 29 (29 x 4 = 116) and of 58 (58 x 2 = 116). Final answer: the smallest possible number is 116.

6. The traffic lights at GPO chowk changes after every 30 seconds, 60 seconds, 90 seconds and 120 seconds respectively. If they change simultaneously at 9 a.m after what time will they change again simultaneously? (Hint: 1 minute = 60 seconds).

Step 1: Find L.C.M of 30, 60, 90 and 120. Step 2: 120 is a multiple of 30 and 60, and L.C.M of 90 and 120 is 360. Step 3: 360 seconds = 6 minutes. Step 4: 9:00 a.m. + 6 minutes = 9:06 a.m. Final answer: they will change together again at 9:06 a.m.

7. Find the smallest 3 digit number divisible by 16, 24 and 30 respectively.

Step 1: Find L.C.M of 16, 24 and 30. Step 2: 16 = 2^4, 24 = 2^3 x 3, 30 = 2 x 3 x 5, so L.C.M = 2^4 x 3 x 5 = 240. Step 3: 240 is already a 3-digit number. Final answer: the smallest 3-digit number divisible by 16, 24 and 30 is 240.

Thinking Time (book page 14): What is the smallest 4-digit number which is divisible by 21, 56 and 63?

Step 1: Find L.C.M of 21, 56 and 63. Step 2: 21 = 3 x 7, 56 = 2^3 x 7, 63 = 3^2 x 7, so L.C.M = 2^3 x 3^2 x 7 = 504. Step 3: Multiples of 504: 504, 1,008, ... Step 4: 504 is 3-digit, so the first 4-digit multiple is 1,008. Final answer: 1,008.

Exercise 1.5 — Relationship between H.C.F and L.C.M (book page 16)

1. Find the relationship between H.C.F and L.C.M of the following numbers: (i) 10 and 25 (ii) 8 and 28 (iii) 34 and 68 (iv) 35 and 77 (v) 110 and 230 (vi) 160 and 380.

Step 1: In every case, H.C.F x L.C.M = product of the two numbers. (i) H.C.F = 5, L.C.M = 50, 5 x 50 = 250 = 10 x 25. (ii) H.C.F = 4, L.C.M = 56, 4 x 56 = 224 = 8 x 28. (iii) H.C.F = 34, L.C.M = 68, 34 x 68 = 2,312 = 34 x 68. (iv) H.C.F =

7, L.C.M = 385, 7 x 385 = 2,695 = 35 x 77. (v) H.C.F = 10, L.C.M = 2,530, 10 x 2,530 = 25,300 = 110 x 230. (vi) H.C.F = 20, L.C.M = 3,040, 20 x 3,040 = 60,800 = 160 x 380. Final answer: for two numbers, H.C.F x L.C.M = product of the numbers.

2. Find the relationship between H.C.F and L.C.M of 680 and 584.

Step 1: H.C.F of 680 and 584 = 8. Step 2: L.C.M of 680 and 584 = 49,640. Step 3: 8 x 49,640 = 397,120. Step 4: 680 x 584 = 397,120. Final answer: H.C.F x L.C.M = product of the numbers, so the relationship holds: 8 x 49,640 = 680 x 584.

3. The H.C.F of two numbers is 36 and their L.C.M is 1260. If one of the numbers is 72, find the other number.

Step 1: Use H.C.F x L.C.M = first number x second number. Step 2: 36 x 1,260 = 45,360. Step 3: Second number = 45,360 / 72 = 630. Final answer: the other number is 630.

4. The H.C.F of two numbers is 48 and their L.C.M is 1440. If one of the numbers is 240, find the other number.

Step 1: 48 x 1,440 = 69,120. Step 2: Other number = 69,120 / 240 = 288. Final answer: the other number is 288.

Challenge (book page 16): Highest common factor (H.C.F) and least common multiple (L.C.M) of two numbers are 10 and 399 respectively. One number is 190, find the other number.

Step 1: H.C.F x L.C.M = 10 x 399 = 3,990. Step 2: Other number = 3,990 / 190 = 21. Final answer: the other number is 21.

Exercise 1.6 — Square numbers (book page 17)

1. Identify the square numbers in the following numbers: (i) 18 (ii) 25 (iii) 17 (iv) 17 (v) 17 (vi) 144 (vii) 79 (viii) 169 (ix) 68 (x) 121.

Step 1: A square number is the product of a number with itself. Step 2: 25 = 5 x 5, 144 = 12 x 12, 169 = 13 x 13, 121 = 11 x 11. Step 3: 18, 17, 79 and 68 are not squares of any whole number. Final answer: the square numbers are 25, 144, 169 and 121. (Note: the numbers 17 appear three times in the book's printed list, as shown.)

2. Calculate the squares of the following numbers: (i) 13 (ii) -8 (iii) 23 (iv) 17 (v) -31 (vi) 43 (vii) -3 (viii) 57 (ix) -71 (x) 93.

Step 1: Multiply each number by itself; a negative number times itself gives a positive result. (i) 13 x 13 = 169. (ii) (-8) x (-8) = 64. (iii) 23 x 23 = 529. (iv) 17 x 17 = 289. (v) (-31) x (-31) = 961. (vi) 43 x 43 = 1,849. (vii) (-3) x (-3) = 9. (viii) 57 x 57 = 3,249. (ix) (-71) x (-71) = 5,041. (x) 93 x 93 = 8,649. Final answers: 169, 64, 529, 289, 961, 1,849, 9, 3,249, 5,041, 8,649.

Review Exercise — Sub-Domain 1 (book pages 17-18)

1(i). A number which divides the dividend completely is called:

factors ✓ **Correct**

multiples

L.C.M

H.C.F

1(ii). Zero (0) is multiple of every number except:

one

two

itself ✓ **Correct**

five

1(iii). Any given number is even if the digit on its ones place is multiple of:

1

2 ✓ **Correct**

3

4

1(iv). Any given number is an odd number if the digit on its ones place is not the multiple of:

1

2 ✓ **Correct**

3

7

1(v). A composite number can always be expressed as a _____ of two primes:

product ✓ **Correct**

difference

sum

none of these

1(vi). A number neither prime nor composite is _____:

1 ✓ **Correct**

2

3

4

1(vii). The product of factors of a given number is always equal to:

prime number

composite number

given number ✓ **Correct**

even number

1(viii). In 5^4:

4 is base and 5 is power

5 and 4 are bases

4 and 5 are powers

5 is base and 4 is power ✓ **Correct**

1(ix). A number which divides all the given numbers is called:

H.C.F ✓ **Correct**

L.C.M

Prime

Composite

1(x). A number which is divisible by all the given numbers is called:

H.C.F

L.C.M ✓ **Correct**

Prime

Composite

2. (i) Define factors and give an example. (ii) Define Multiples and give an example. (iii) What are prime numbers? (iv) What are composite numbers? (v) What do you know about index notation? (vi) Define prime factorization. (vii) Define H.C.F (Highest Common Factor). (viii) Define L.C.M (Least Common Multiple).

Step 1: (i) A factor is a number which divides the dividend completely leaving no remainder, for example 3 is a factor of 42 because 42 / 3 = 14 with remainder 0. (ii) A multiple of a given number is the dividend for which the given number is its factor, for example the multiples of 3 are 3, 6, 9, 12, 15, ... (iii) A prime number is a natural number with only two different factors, 1 and the number itself, for example 2, 3, 5, 7, 11. (iv) A composite number has more than two different factors, for example 4, 6, 8, 9. (v) Index notation is the representation of prime factors in exponential form, for example 36 = 2^2 x 3^2. (vi) Prime factorization is the process of factorizing a number into its prime factors. (vii) H.C.F of two or more numbers is the highest number which

divides all the given numbers. (viii) L.C.M is the least number which is divisible by all the given numbers.

3. For an admission test the number of participants in grade VI, grade VII and grade VIII are 60, 84 and 108. Find the minimum number of participants who can sit in a row.

Step 1: The row size must divide 60, 84 and 108, and the question asks for the minimum number who can sit in a row in the sense of the greatest common row size. Step 2: H.C.F of 60, 84 and 108: 60 = 2^2 x 3 x 5, 84 = 2^2 x 3 x 7, 108 = 2^2 x 3^3, so H.C.F = 2^2 x 3 = 12. Final answer: 12 participants can sit in each row.

4. Three local taxies A, B and C arrive at a stop. Taxi A arrives at the stop every 20 minutes, taxi B arrives every 30 minutes and taxi C arrives every 40 minutes. All three taxies arrive at the stop at 10:30 am altogether. At what time will the three taxies arrive at the stop again?

Step 1: Find L.C.M of 20, 30 and 40. Step 2: 20 = 2^2 x 5, 30 = 2 x 3 x 5, 40 = 2^3 x 5, so L.C.M = 2^3 x 3 x 5 = 120 minutes. Step 3: 120 minutes = 2 hours. Step 4: 10:30 a.m. + 2 hours = 12:30 p.m. Final answer: the three taxies will arrive together again at 12:30 p.m.

5. Find the largest number which when divided by 24, 36 and 60 leaves 3 as remainder in each case.

Step 1: A number that leaves remainder 3 on division by 24, 36 and 60 is 3 more than a common multiple of 24, 36 and 60. Step 2: L.C.M of 24, 36 and 60 = 360. Step 3: Numbers of the form 360k + 3 are 363, 723, ... Step 4: As printed, no upper limit is given, so there is no largest such number; the smallest such number greater than 3 is 363. Final answer: the least number is 363, and no largest number exists without an upper limit.

6. Three bells toll at regular interval of 10 minutes, 20 minutes and 30 minutes respectively. If they toll altogether at 12:00 noon. At what time will they next toll together?

Step 1: Find L.C.M of 10, 20 and 30. Step 2: L.C.M = 60 minutes. Step 3: 12:00 noon + 60 minutes = 1:00 p.m. Final answer: they will next toll together at 1:00 p.m.